Asmit Bhandari

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ACADEMIC DETAILS

Institute	Degree	Score	Duration
Graphic Era Hill University, Dehradun	B.Tech Computer Science Engineering	CGPA: 7.15/10	Aug 2023 – Present
Jawahar Navodaya Vidyalaya, Rudraprayag	CBSE Class XII	76.60%	2022
Jawahar Navodaya Vidyalaya, Rudraprayag	CBSE Class X	92.60%	2020

INTERNSHIPS

AI-assisted Full Stack Development Intern — **Jun 2026 – Present**
TBI Graphic Era University — *Dehradun*

- **Technologies:** React.js, TypeScript, FastAPI, Python, Tailwind CSS, MongoDB Atlas, RESTful APIs, Render, Git
- Developed **VPRO**, maritime analytics platform integrating **15+ RESTful API endpoints** for dashboard visualization, report management, AI-assisted workflows, and vessel performance analytics.
- Built **15+ reusable React components** for responsive dashboards, AI-assisted workflows, interactive data visualization, and real-time maritime analytics.
- **Live Project:** vpro.asmitlabs.me

Freelance Web Developer | Bhuyogmantra — **Jul 2025 – Aug 2025**

- **Technologies:** React.js, TypeScript, Supabase, Tailwind CSS, RESTful APIs, Git, Vercel
- Rebuilt a legacy static website into a modern full-stack platform featuring **6 responsive pages**, Supabase integration, SEO optimization, and deployment, serving approximately **100 monthly users**.
- **Live Project:** bhuyogmantra.com

PROJECTS

CodeMorpher | *React.js, FastAPI, Java, ANTLR4, React Flow, Compiler Design, Abstract Syntax Tree (AST), Intermediate Representation (IR), Multithreading* — **Mar 2026 – May 2026**

- Engineered a source-to-source transpiler supporting **4 target programming languages** (C, C++, Java, and JavaScript) through lexical analysis, parsing, Abstract Syntax Tree (AST) generation, Intermediate Representation (IR), and interactive compiler pipeline visualization using React Flow.
- Implemented **parallel processing using multithreading** to execute independent compiler stages concurrently, reducing overall transpilation time by approximately **15–25%** while improving application responsiveness and processing efficiency.

AI Voice Detection System | *Python, FastAPI, Transformers, Wav2Vec2, Hugging Face, Librosa, RESTful APIs* — **Jan 2026 – Feb 2026**

- Designed and deployed a Transformer-based voice detection system using a fine-tuned Wav2Vec2 model for real-time classification of AI-generated and human speech.
- Achieved **89.1% AI detection accuracy**, **85.8% human detection accuracy**, and **62.5% zero-shot multilingual accuracy** across Hindi, Tamil, Telugu, and Malayalam while exposing the model through a production-ready FastAPI service with explainable confidence scoring.

Braille Bridge | *React.js, FastAPI, PostgreSQL, SQLAlchemy, JWT, Tesseract OCR, Liblouis, Google Cloud Text-to-Speech, HTML5, CSS3, JavaScript* — **Oct 2025 – Nov 2025**

- Architected an accessibility platform supporting **12 Indian languages** with multilingual Voice-to-Braille and Braille-to-Voice conversion using OCR, Braille translation, and text-to-speech technologies.
- Built a secure FastAPI backend with **15+ RESTful API endpoints**, JWT authentication, document processing, translation management, and OCR-based extraction for PDF and image-based educational materials.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, JavaScript
Frontend: React.js, Tailwind CSS, Bootstrap, HTML5, CSS3
Backend / Databases: Node.js, Flask, MySQL, PostgreSQL, Oracle, MongoDB Atlas, Supabase
Machine Learning: NumPy, Pandas, Scikit-learn, TensorFlow
APIs: REST APIs, Firebase APIs, OpenAPI (Swagger), Cloud APIs
Developer Tools: Git, GitHub, VS Code, Postman, Jupyter Notebook, Render, Vercel, Netlify, Google Cloud, Canva
Core Concepts: Object-Oriented Programming, System Design, Data Structures, Algorithms, DBMS, Operating Systems

CERTIFICATES

- **Foundation of Google Cloud Computing** — NPTEL (Feb – Apr 2025)
- **Data Analytics with Python** — NPTEL (Jul – Sep 2024)